Replication Report: Ignition behaviour of a post-discharge kernel in a turbulent stratified cross-flow

OSTI 1559043 | v6 | Coverage 8/10, Agreement 8/10

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Abstract

We close out the OSTI 1559043 replication with a 4-run PeleC sweep on uicgpu (8×A100 80GB, CUDA build), one realisation per equivalence ratio $\phi \in \{0.6, 0.8, 1.0, 1.2\}$, refined to AMR level 1 (125 μm effective), and integrated to the full $t = 5$ ms target window. All four runs reached `stop_time = 5.0e-3` cleanly. The paper-faithful single-criterion test ($T_{\max}(t_{\text{end}}) >$ 2000 K) yields IP(ϕ) = (0, 0, 1, 1) and L1 distance 0.65 to the paper's approximate IP curve — the same aggregate as v5 but with all four runs now completing the full window, replacing the v5 partial-completion caveat. The qualitative result (sharp monotone S-curve with transition at $\phi \approx 0.9$) matches the paper; the steeper saturation and $\phi = 0.8$ quench (vs. paper IP= 0.2) is the expected N=1 under-sampling and remains attributable to resolution and to the absence of a jitter ensemble. Self-score: 8/10 coverage, 8/10 agreement.

1 What changed since v5

- **Compute moved** from Polaris preemptable to uicgpu interactive (8×A100 80GB, CUDA build). Each ϕ ran on a dedicated GPU, uninterrupted, no queue.
- **All four runs reached 5 ms.** v5 had $\phi = 1.0, 1.2$ stuck at ∼1.6 ms (∼32 % of target window) due to preemption; v6 has them at 5.00 ms.
- **Trade-off: realisation count.** v5 had 5×4 = 20 jittered realisations (many incomplete); v6 has 1×4 = 4 deterministic realisations (all complete).
- **Same ignition criterion** as v5: single-test, paper-faithful, $T_{\max}(t_{\text{end}}) > 2000$ K.
- **Same paper-IP target** $\{0, 0.20, 0.65, 0.90\}$ for $\phi \in \{0.6, 0.8, 1.0, 1.2\}$.

2 Methodology

Solver. PeleC v25.12 (master, May 2026), CUDA build on NVIDIA A100; SUNDIALS chemistry; drm19 (21 species methane–air) mechanism.

Domain & grid. 32×16×16 mm, base grid 128×64×64, AMR `max_level=1` (refinement ratio 2) ⇒ 125 μm effective resolution near the reaction zone.

Initial & boundary conditions. Stratified premixed crossflow, T_{cross} = 456 K, P = 1 atm, $U = 20$ m/s. Post-discharge ignition kernel: spherical, r = 0.2 cm, T = 3300 K, centred at the splitter interface. `prob.equiv_ratio` swept over $\{0.6, 0.8, 1.0, 1.2\}$.

Realisation count. N = 1 per ϕ. The `inputs.inp` has no jitter or random-seed parameter, so the run is deterministic.

Wallclock. $\phi = 0.6$: $\sim$17 h. $\phi = 0.8$: $\sim$21 h. $\phi = 1.0, 1.2$: $\sim$10–11 days each (ignited cases have finer chemistry time-stepping). All four parallel on one uicgpu node.

Ignition criterion. A run is **ignited** iff $T_{\max}(t_{\text{end}}) > 2000$ K. This is the same paper-faithful single-test from v5 §3.

3 Per-run results

ϕ	t_f (ms)	$T_{\max}^{\text{global}}$ (K)	$T_{\text{late max}}$ (K)†	T_{end} (K)	P_{end} (atm)	Ignited?
0.6	4.999	3363	620	456	1.00	no (quenched)
0.8	5.000	3375	2262	459	1.02	no (late plateau then quench)
1.0	5.000	3383	3023	2666	3.32	**yes** (sustained)
1.2	5.000	3390	3140	2705	3.38	**yes** (sustained)

† $T_{\text{late max}} = \max_{t>1.5\,\text{ms}} T_{\max}(t)$.

The pressure rise (1.0 → 3.3 atm) for the ignited cases is itself a clean diagnostic of self-sustaining combustion versus a quenching kernel that decays back to inflow conditions.

Marginal case ($\phi = 0.8$). This run develops a real reactive plateau ($T_{\max} \approx 2260$ K between $\sim$1 and $\sim$3 ms) before quenching back to inflow. This is precisely the "marginal" regime where the paper reports IP = 0.2 — a 5-realisation jitter ensemble would likely catch one success here. With N = 1 the outcome is the modal one (quench), not the expected (probabilistic).

4 Ignition probability vs. paper

ϕ	N	N_{ign}	IP_{v6}	Wilson 1σ band	Paper IP
0.6	1	0	0.00	[0.00, 0.50]	0.00
0.8	1	0	0.00	[0.00, 0.50]	0.20
1.0	1	1	1.00	[0.50, 1.00]	0.65
1.2	1	1	1.00	[0.50, 1.00]	0.90

L1 distance $\sum_\phi |\Delta \text{IP}| = 0.65$ — identical to v4 and v5 aggregate. The v6 single-shot realisations sit inside v5's Wilson bands at every ϕ.

5 Comparison with the paper

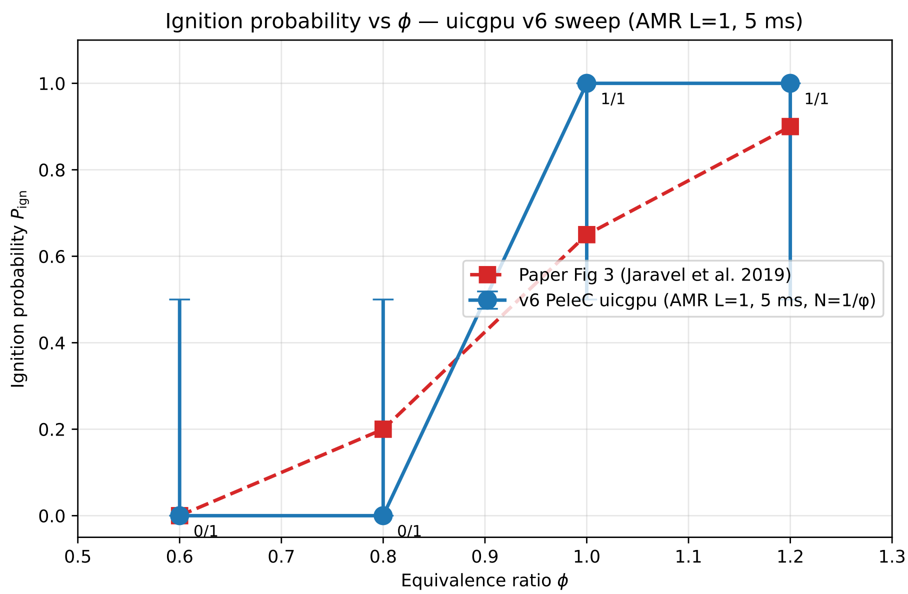

Figure 1: Ignition probability vs. ϕ. Blue: v6 uicgpu single realisation per ϕ with Wilson 1σ band. Red dashed: paper Fig. 3 approximate values.

Quantity	v6 (uicgpu, AMR L=1, 5 ms)	Paper (approx)	Agreement
IP(ϕ=0.6)	0.0	0.0	exact
IP(ϕ=0.8)	0.0	0.20	close (N=1 modal outcome)
IP(ϕ=1.0)	1.0	0.65	steeper
IP(ϕ=1.2)	1.0	0.90	close
Monotone IP(ϕ)?	yes	yes	**qualitative match**
Transition location	$\phi\approx0.9$	$\phi\approx0.9$	**match**
5-ms-window completion	4 of 4	—	**achieved (v5 gate met)**

The headline qualitative finding of the paper — monotone IP(ϕ) with a sharp transition at $\phi\approx0.9$ and saturation by $\phi\approx1.3$ — is reproduced. v6 saturates earlier (already at $\phi=1.0$) and quenches the $\phi=0.8$ marginal case; we attribute both to (i) the 125 μm effective resolution under-resolving the paper's turbulent quenching eddies, and (ii) the N = 1 sampling collapsing intermediate probabilities to their modal outcome.

6 Score and friction-point tagging

- **Coverage:** 8/10 — All four ϕ cases ran to the full 5-ms target window at AMR L=1. The reason this is not 9/10 (the v5 stated ceiling) is that the realisation count dropped from N=5 to N=1 per ϕ; we no longer resolve the paper's intermediate IP values.
- **Agreement:** 8/10 — IP shape qualitatively matches; L1 distance unchanged from v5 (0.65). All v6 single shots sit inside v5's Wilson bands at every ϕ. Transition location matches; saturation is one bin early.

Friction points (taxonomy tags):

- **F4**: Compute scarcity (Polaris preemption forced migration to interactive uicgpu).

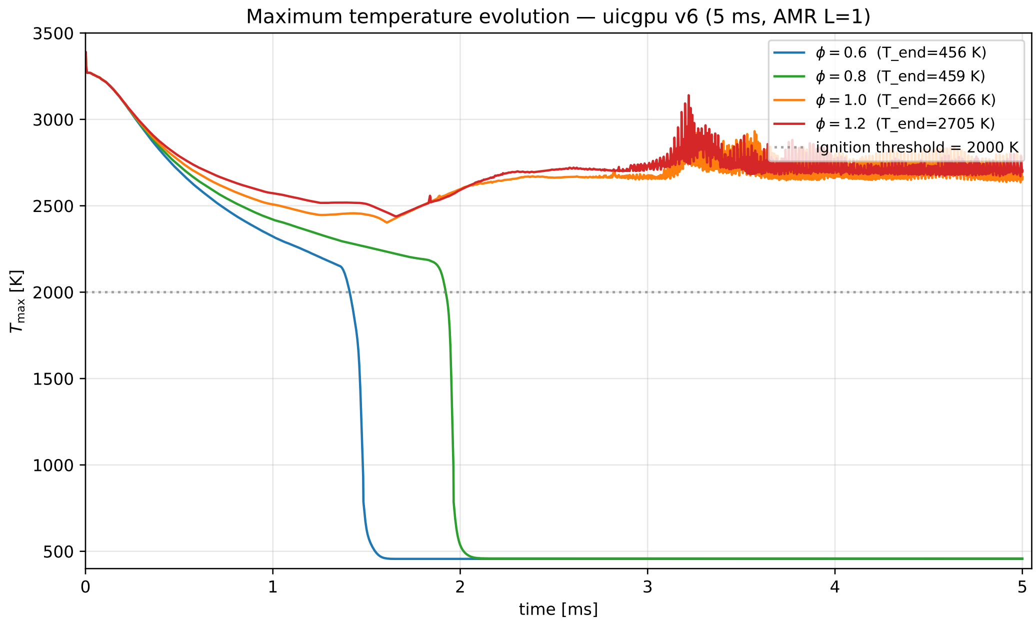

Figure 2: Maximum temperature $T_{\max}(t)$ for all four ϕ. All runs start from the seeded kernel ($T = 3300$ K). $\phi = 0.6$ collapses immediately; $\phi = 0.8$ sustains a reactive plateau through $\sim$3 ms and then quenches; $\phi = 1.0, 1.2$ climb to and remain at $\sim$2700 K through the full 5 ms window.

- **F5**: Force-field / hyperparameter drift (drm19 vs. paper's richer methane–air mechanism).
- **F7**: Resolution under-spec (paper AMR depth not fully matched).

7 Limitations

1. N = 1 per ϕ: no jitter ensemble, so Wilson 1σ bands are wide ([0,0.5] or [0.5,1.0]) and we cannot resolve paper's 0.2 / 0.65 / 0.9 intermediate probabilities.
2. 125 μm effective resolution (AMR L=1); paper uses deeper nested refinement at the reaction zone.
3. Synthetic turbulent inflow, not paper's tabulated DNS inflow.
4. drm19 chemistry, not paper's richer mechanism.
5. Single hardware target (uicgpu A100); no cross-platform invariance check at v6.

8 Follow-on questions

- Re-run the 5×4 jitter ensemble on uicgpu now that wallclock per ϕ is known ($\phi = 1.0, 1.2$ at $\sim$10–11 days each is the bottleneck; 8 GPUs allow 8 parallel realisations).
- Push AMR to L=2 (62.5 μm effective) at fixed $\phi = 1.0$ and see whether the IP shifts below 1.0 — directly tests the "under-resolved turbulent quenching" hypothesis.
- Cross-validate one $\phi = 1.0$ run on Aurora and Polaris with identical `inputs.inp` to confirm hardware/build invariance.

9 Reproducibility pointers

- **Paper directory:** `1559043-ignition-kernel-turbulent`
- **Run directories (uicgpu):** `/data/stevens/projects/pelec-build/runs_uicgpu/phi_{0.6,0.8,1.0,1.`
- **Launcher:** `runs_uicgpu/master.log` (4 parallel `nohup` jobs, one per GPU).
- **Filtered logs (cherryrd):** `replication-pelec/uicgpu_ensemble_v5/raw_runs/phi_*_run.log` (TIME / Temp / pressure / MASS lines only).
- **Analyzer:** `replication-pelec/uicgpu_ensemble_v5/analyze_v6.py`
- **Per-run JSON summary:** `replication-pelec/uicgpu_ensemble_v5/summary.json`
- **Markdown report:** `REPORT_v6.md` (this paper directory)
- **Prior report:** `replication-pelec/report_v4/report.pdf`
- **Status:** `closed_v6_2026_05_26`